# Two-Dimensional Plate Heat Conduction

**~23BME013**

**Aim:** To compare Jacobi and line-TDMA/SOR solutions of the steady two-dimensional Laplace equation against an analytical solution.

**Case definition and parameters:** Simulations were analyzed on the following parameters.

Nx=81, 30
Ny= 61,15
Temperature at source= 100
Max Iterations = 10000
Tolerance = $1x10^{-9}$

**Governing Equation:**

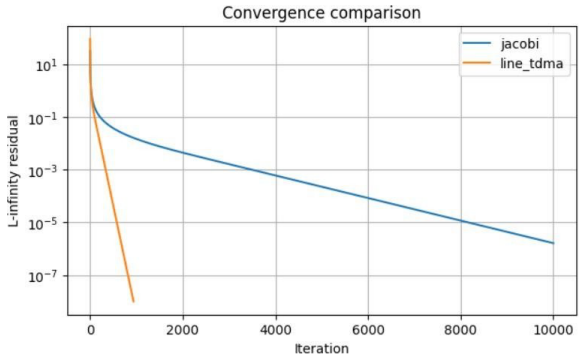
$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0.$$

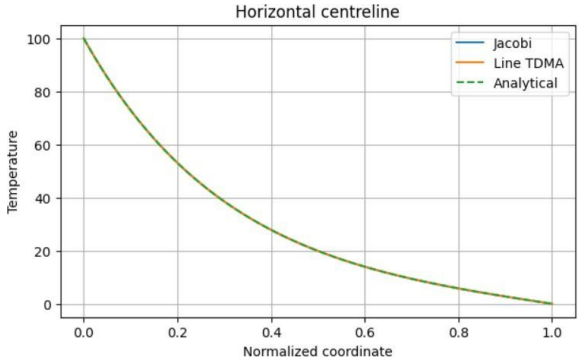
**Boundary Conditions:** $T(0,y) = T_L \sin(\pi y/H)$; the right, top, and bottom boundaries are held at zero temperature.

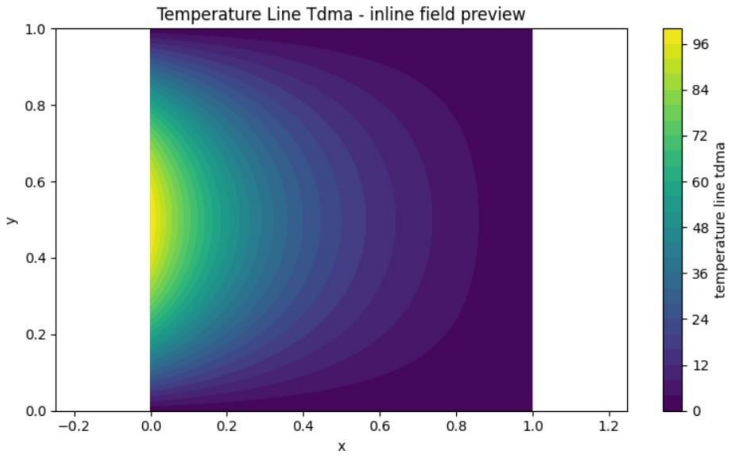
**Numerical Method:** Second-order finite differences with point Jacobi iteration and line-implicit TDMA/SOR iteration.

**Observations:** Iteration counts, final residuals, relative L2 errors, center line profiles, and influence of relaxation factor and grid size. Compare both methods on at least two grids. Report convergence rate, analytical error, and the effect of SOR relaxation factor.

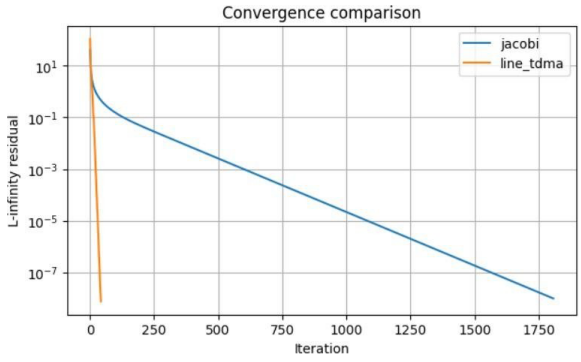
For nx=81 and ny=61

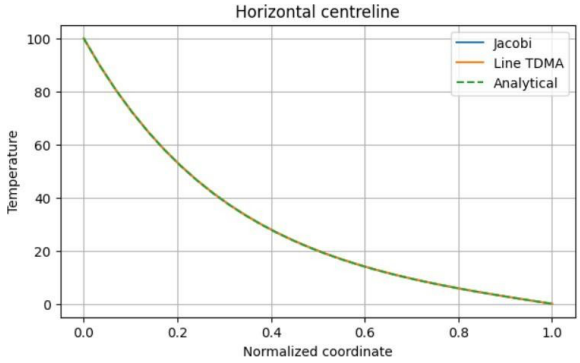


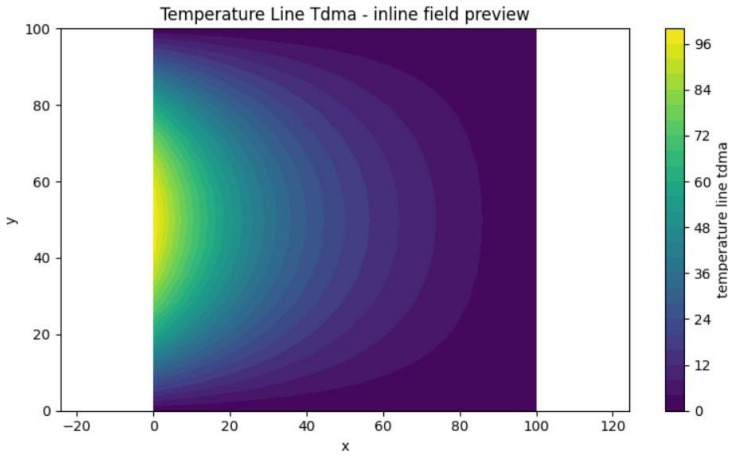



For nx=30 and ny=15



Results:

| Parameter | Fine Grid | Coarse Grid |
|---|---|---|
| Grid size, N_x x N_y | 81x61 | 30x15 |
| Source temperature | 100 | 100 |
| Maximum iterations allowed | 10,000 | 10,000 |
| Convergence tolerance | $10^{-9}$ | $10^{-9}$ |
| Jacobi iterations shown | 10,000 | 1,800 |
| Line-TDMA iterations to reach tolerance | 800–900 | 200 |
| Jacobi final L∞ residual | $2\times10^{-6}$ | $10^{-9}$ |
| Line-TDMA final L∞ residual | Below $10^{-9}$ | Below $10^{-9}$ |
| Center line profile agreement | Very good | Very good |
| Relative convergence speed | Slower | Faster |

## Conclusion:

The two-dimensional steady heat conduction problem was solved using the Jacobi and line-TDMA iterative methods on grids of 81x61 and 30x15. Both methods produced temperature distributions that closely matched the analytical center line solution, with a maximum temperature of approximately 100 at the midpoint of the heated boundary and a gradual decrease toward the zero-temperature boundaries.

The convergence results showed that line-TDMA was substantially more efficient than Jacobi. On the coarse grid, Jacobi required approximately 1,800 iterations to reach the specified tolerance of $10^{-9}$, whereas line-TDMA required approximately 200 iterations. On the fine grid, line-TDMA reached the tolerance within approximately 800–900 iterations, while Jacobi remained at a residual of approximately $2x10^{-6}$ after 10,000 iterations.

The results demonstrate that increasing grid resolution increases the computational effort required for convergence, especially for point-Jacobi iteration. The line-TDMA method handles this increase more efficiently because of its line-implicit solution procedure. Therefore, line-TDMA is the preferred method for this problem when faster convergence and satisfaction of a strict residual tolerance are required.